Physics-Informed Machine Learning Framework for Predicting Micro-viscoelastic and Micro-structural Properties of Granular Composites

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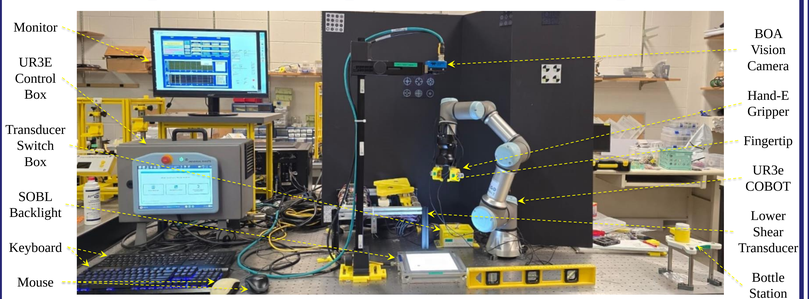
Problem Statement

- Pharmaceutical Continuous Manufacturing (CM) currently lacks reliable, real-time techniques to monitor tablet microstructure and micro-viscoelastic properties that govern critical quality attributes and drug-release performance.
- Although ultrasonic testing provides a non-destructive method suitable for Real-Time Release Testing (RTRT), accurate material characterization remains challenging due to complex inverse wave propagation problem.
- The absence of a reliable and interpretable approach to address the ultrasonic inverse problem limits the effective use of ultrasonic methods for RTRT.

Main Project Objective

- Develop a Physics-Informed Machine Learning (PIML) model for the material parameters characterization of micro-viscoelastic granular composite, with a focus on pharmaceutical Oral Solid Dosage (OSD) forms to monitor real-time critical quality attributes in pharmaceutical CM.

TOTO Experimental Rig for Waveform Acquisition



Mathematical Framework for Characterization Work

Wave equation for the first-order visco-elastic Standard Linear Solid (SLS) material model, also known as Zener visco-elastic model:

$$\frac{\partial^2 u}{\partial t^2} + p_1 \frac{\partial^3 u}{\partial t^3} = \frac{E}{\rho}\left(\frac{\partial^2 u}{\partial x^2} + q_1 \frac{\partial^3 u}{\partial x^2 \partial t}\right)$$

The dispersion relation of the visco-elastic attenuation is obtained using the dispersive displacement field of the propagating pulse

$$u(x,t) = U(\omega)\, e^{i(\kappa_v(\omega)x - \omega t)} \text{ with } \kappa_L(\omega) = \beta_L(\omega) + i\,\alpha_L(\omega)$$

Complex wavenumber equation for the material attributes:

$$\kappa_{\mathrm{L}}(\omega) = \frac{1}{ih} ln\left(\frac{\mathcal{F}\{u(x=h,t)\}}{\mathcal{F}\{u(x=0,t)\}}\right) = \beta_L(\omega) + i\alpha_L(\omega),$$

where $\mathcal{F}\{u(x=h,t)\}$ and $\mathcal{F}\{u(x=0,t)\}$ are the FFT of experimentally acquired waveforms. Phase c_p and group c_g velocities are obtained as:

$c_p(\omega) = \omega/\beta_{\mathrm{L}}(\omega)$ and
$c_g(\omega) = d\omega/d\beta_{\mathrm{L}}(\omega) = 1/(d\beta_{\mathrm{L}}(\omega)/d\omega)$.

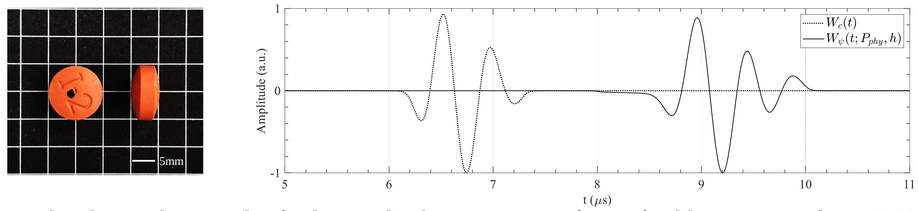
At $\omega = 0$, there is no dynamic effect in play (static), meaning no effect of viscoelasticity is present. Assuming real wave number curve $\beta_{\mathrm{L}}(\omega)$ by a cubic polynomial, so it can be written as:

$$\beta_{\mathrm{L}}(\omega) = a\,\omega^3 + b\,\omega^2 + \frac{1}{c_L}\,\omega + 0 \;\Rightarrow c_g(0) = 1/(d\beta_{\mathrm{L}}(\omega)/d\omega)|_{\omega=0} = c_L$$

Scattering models for composite solids with micro-structures are as follows:

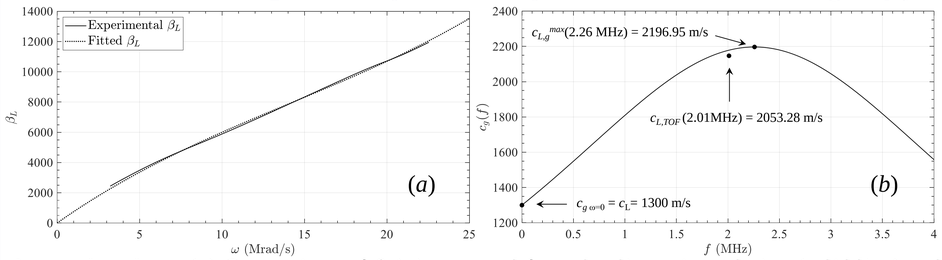
Rayleigh Regime $\lambda \simeq 2\pi d \gg d \Rightarrow \alpha^s(f,d) = c_r d^3 f^4 = c_r d^3/\lambda^4$
Stochastic Regime $\lambda \approx d \Rightarrow \alpha^s(f,d) = c_s d f^2 = c_s d/\lambda^2$
Diffusion Regime $\lambda \ll d \Rightarrow \alpha^s(d) = c_d/d$

Acquired Waveforms of a Pharmaceutical Tablet



Normalized time-domain plot for longitudinal pressure waveform of calibration waveform $W_c(t)$ (dashed line) and tablet's waveform $W_\psi(t; P_{phy}, h)$ (solid line) containing the primary pulse acquired in TOTO experimental rig in PCM mode.

Extraction of Group Velocity with Polynomial Curve Fitting of β_L



Approximations: (*a*) Comparison of $\beta_L(\omega)$ extracted from the dispersion relation (solid line) and its polynomial fit (dashed line) in the angular-frequency domain. (*b*) Group velocity $c_{L,g}$ obtained from the polynomial fit of $\beta_L(\omega)$ showing the $c_{L,g,max}$ and $c_{L,ToF}$. The relative discrepancy is 6.96%.

Variational Procedure for the Implementation of PIML Framework

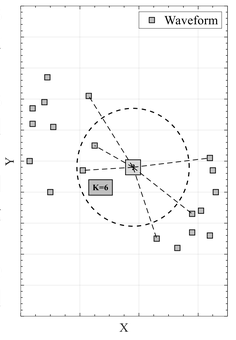
Step 1: Acquire ultrasonic waveforms (i) Calibration $W_c(t)$ (ii) Sample $W_\psi(t; P_{phy}, h)$
Step 2: Estimate the initial MA, $P_0\,(p_1, q_1, c_1, c_{L,g})$ using the *WavePropagator*
Step 3: $P_j \rightarrow P_0 + j\,\Delta P_0$ for the discretization space of j = -*n*, … -2, -1, 0, 1, 2, 3 ..., *n*
Step 4: Generate a set of synthetic waveforms $W_\psi(t;P_j, h)$ for the discretized parameter space
Step 5: $\|e(j)\|$; determine the error, *e(j) for the set of synthetic waveforms* $W_v(t;P_j, h)$
Step 6: *j* giving min ($\|e(j)\|$) will be the new P_j^{opt}
Step 7: Train, test and validate the *WaveML28* model using the training waveform set of $W_v(t; P_j, h)$, generated based on P_j^{popt} and extract the P_{ML} for sample waveform $W_\psi(t; P_{phy}, h)$
Step 8: Train, test and validate the *DispersionML57* model in the ω-$k(\omega)$ domain and extract P_{DML} for sample waveform $W_\psi(t; P_{phy}, h)$
Step 9: ($\|e_{DC}(j)\|$) in the ω-$k(\omega)$ domain using the difference between $k_\psi(\omega)$ and $k_j(\omega; P_j)$.
Step 10: Compare the computational efficiency of all variational method.

Discretized Design Space for Synthetic Training Dataset Generation

Material Attributes (MA)	Unit	Design Space of the Training Data Set of p_j, q_j, c_j and $c_{L,g,j}$					Generated Synthetic Waveforms
		Minimum	Optimal	Maximum	Number of Steps	Step Size	
		$P_{min} \rightarrow P_{popt} - j\Delta p_j$	P_{popt}	$P_{min} \rightarrow P_{popt} + j\Delta p_j$	*n*	Δp_j	
p_1	μs	0.41	0.66	0.91	11	0.05	12,100
q_1	μs	1.4	1.65	1.9	11	0.05	
c_1	$\times 10^{-27}$ (s/m)^4	0.31	1.81	4.81	10	0.5	
$c_{L,g}$	m/s	1250	1300	1340	10	10	

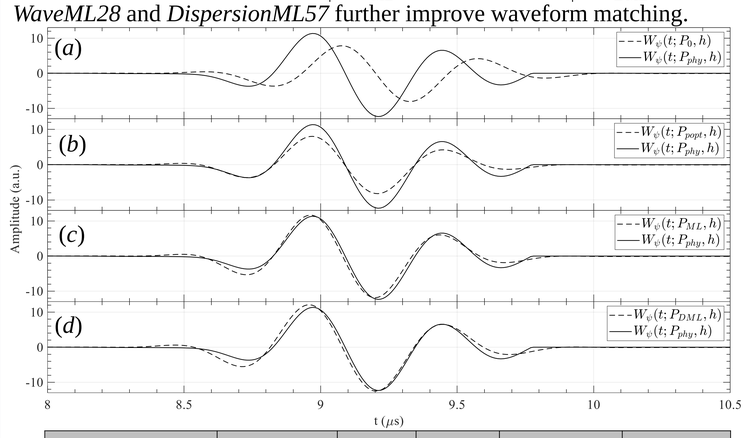
K-Nearest Neighbors (KNN) Regression Framework

- K-Nearest Neighbors (KNN) regression is employed as a physics-informed ML framework for *WaveML28* and *DispersionML57*.
- Each synthetic waveform $W_\psi(t; P_j, h)$ is represented as a point in a feature space and used as training data X_{train}.
- For a given test waveform X_{test}, distance-based similarity metrics are computed to identify the *k* nearest neighbors.
- MA $P(p_1, q_1, c_1, c_{L,g})$ associated with these neighbors are combined using weighted averaging to predict the output parameters Y.



Key Results: Predicted Micro-Viscoelastic Properties

Progressive waveform reconstruction across the variational PIML steps is shown. (*a*) $W_\psi(t;P_0,h)$ exhibit noticeable mismatch, while (*b*) the $W_\psi(t;P_{popt},h)$ improve *TOA* and first-phase agreement. Reconstructions using KNN-predicted (*c*) $W_\psi(t;P_{ML},h)$ and (*d*) $W_\psi(t;P_{DML},h)$ from *WaveML28* and *DispersionML57* further improve waveform matching.



Sample Set	Parameter Source	p_1 (μs)	q_1 (μs)	$c_1 \times 10^{-27}$ $(s/m)^4$	$c_{L,g}$ (m/s)
LT_TOTO_P_01	P_{DML}	0.76	1.85	2.36	1321
	P_{ML}	0.63	1.60	1.51	1290
	P_{popt}	0.66	1.65	1.81	1300
	P_0	0.71	1.62	2.87	1300

Conclusions and Remarks

- The ML-driven modeling framework for the *WaveML28* and *DispersionML57* accurately extracts visco-elastic and scattering properties from ultrasonic responses of the *pharmaceutical* tablets.
- Residual mismatches observed in the early-time response and waveform tail are attributed to limitations of the current *WavePropagator* physics model and/or unmodeled experimental effects.
- The ML approach's potential in RTR testing with ultrasound in pharmaceutical Continuous Manufacturing (CM) is demonstrated.

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